Segurança de Redes

Network Security

Mestrado em Cibersegurança (MCS)

Mestrado em Redes de Computadores (MRC)

Departamento de Informática, Universidade do Minho

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1° Semestre (26/27)

Adapted from Prof. Paulo Carvalho's (Dept. Informática – UMinho) slides, which were originally based on:
J.F Kurose and K.W. Ross, "Computer Networking: A Top-Down Approach, 8th ed., Pearson, 2020. Teaching material.

Security at Network Layer



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Network Layer: Virtual Private Networks

def Virtual Private Network

A VPN creates a secure, encrypted connection between a user's device and a remote server or network over the Internet

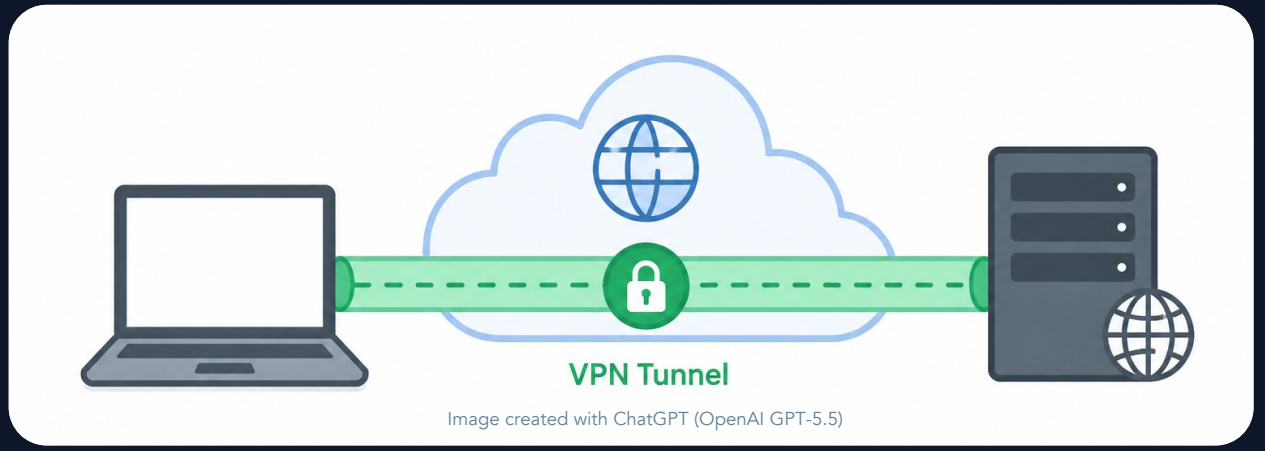
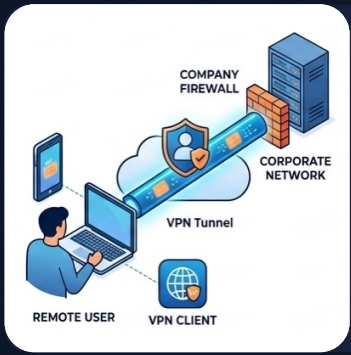


Image created with ChatGPT (OpenAI GPT-5.5)

- Key goals: CIA (Confidentiality, Integrity, Availability), anonymity, access control
- Operation (Tunneling)
  - VPNs encapsulate data inside another protocol (e.g., IPsec ESP, TLS) and add an outer IP header, allowing the data to traverse public networks securely
  - Full tunneling: all user traffic is sent through the VPN
  - Split tunneling: only selected traffic (e.g., corporate resources) passes through the VPN, while the rest uses the public network directly

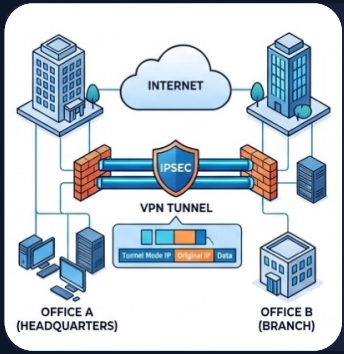
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Network Layer: Types of VPNs (1/2)



Remote Access VPN

- involves a VPN client that connects the user's device to a VPN server on the corporate network
- protects data transmitted between the user's device and the company network



Site-to-Site VPN

- connects entire networks (e.g., two or more office locations) securely over the Internet
- either Intranet-based or Extranet-based Site-to-Site VPN
- often implemented using IPsec in tunnel mode

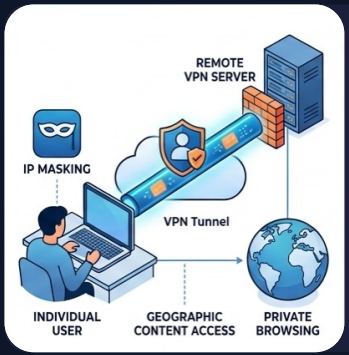


TLS VPN

- commonly used in web-based applications (no additional client sw required)
- uses TLS protocols to encrypt traffic, allowing users to connect through their browsers; IPsec-based VPNs offer greater privacy (in ESP tunnel mode)

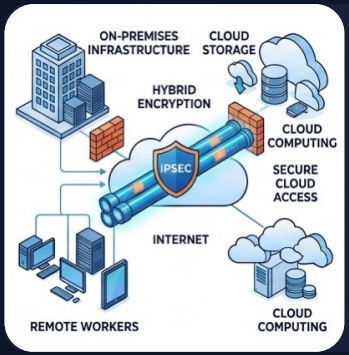
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Network Layer: Types of VPNs (2/2)



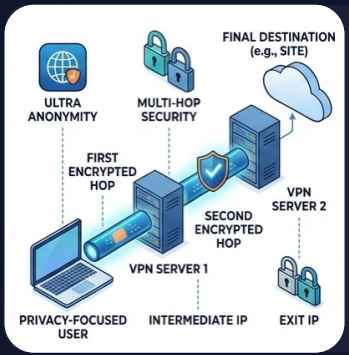
Personal VPN

- used by individuals for privacy, anonymity, and bypassing geographic restrictions (e.g., accessing content that is restricted in certain countries)
- routes user's traffic through a remote VPN server, masking original IP address
- **privacy does not mean complete anonymity**: websites can still identify users through logins, cookies or browser fingerprinting



Cloud VPN

- crucial way to secure communication between on-premises infrastructure and cloud services
- helps org to protect data in hybrid environments, and secure remote work in the cloud
- often follows the **site-to-site model**, with a gateway at the organisation and another in the cloud.



Double VPNs

- often used in privacy-conscious settings, route traffic through two VPN servers, providing an extra layer of anonymity and security
- address privacy-focused use cases and help mitigating threats such as traffic correlation attacks
- **trade-off**: the extra server and processing usually increase latency and may reduce throughput

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Network Layer: Open-Source VPN protocols

VPN Protocol	Security Strength	Speed	Overhead	Known Vulnerabilities	Key Features
OpenVPN (L7/L4)	Strong (AES-256, RSA, HMAC, TLS)	Moderate	High (TCP/UDP, encryption overhead)	No major vulnerabilities but requires proper config	Multiple platforms, customizable, highly secure, flexible with TCP/UDP
WireGuard (L3)	Strong (ChaCha20, Curve25519, Poly1305)	High	Low (lean codebase, smaller attack surface)	Early versions had protocol-level issues, now secure	Simplicity, very fast, minimal code, ideal for modern use cases
IKEv2/IPsec (L3)	Strong (AES-256 or ChaCha20, RSA, or ECDH, SHA-2)	High	Moderate (efficient for mobile)	Resistant to most known attacks if correctly config	Stable and secure for mobile devices, supports mobility, multihoming, fast reconnects
L2TP/IPsec (L2,L3)	Moderate (IPsec security)	Moderate	High	Vulnerable to MITM attacks without proper IPsec config	Older but still used in legacy systems
SoftEther (L2,L7/L4) (L3 optional)	Strong (SSL/TLS)	Moderate	Moderate	No major vulnerabilities known	Multi-protocol support (L2TP, OpenVPN, SSTP, etc.), easy to set up

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Security at Network Layer

IPv6 Security



Image created with ChatGPT (OpenAI GPT-5.5)

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Network Layer: IPv6 security

- IPv6 introduces design changes that affect network security, but still requires appropriate security controls
  - one of its focus: be resilient to network attacks
- Main security-related features:
  - Expanded address space
    - from 2^{32} to 2^{128}
    - limits scanning and spoofing attacks, making random address guessing of active hosts impractical
  - IPsec support (with AH and ESP)
    - initially mandatory, currently recommended (RFC 6434) (extension headers)
  - Simplified header structure
    - eases header inspection; decreases attack surface
  - No broadcast

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Network Layer: IPv6 security

- Neighbor Discovery Protocol (NDP)
  - replaces ARP with NDP, which uses multicast instead of broadcast
  - to counter NDP spoofing, adds a security extension (SEcure ND) for cryptographic address verification
  - SEND uses Cryptographically Generated Addresses (CGAs)
  - CGAs link an IPv6 address to a public key, verifying that the source is the legitimate owner of that address
- Built-in privacy extensions (RFC 8981)
  - for temporary, random IP address assignment, reducing the risk of user tracking and identification by rotating device addresses
- Removal of NAT
  - because IPv6 provides a massive supply of unique global addresses, devices do not need to share a single public IP address the way they do with IPv4
  - reduces attack surface
- Fragmentation
  - intermediate routers cannot chop packets into pieces
  - no fragmentation by default; restricted to source (If a packet is too large for a router along the path, the router drops it and sends an ICMPv6 "Packet Too Big" message back to the source)

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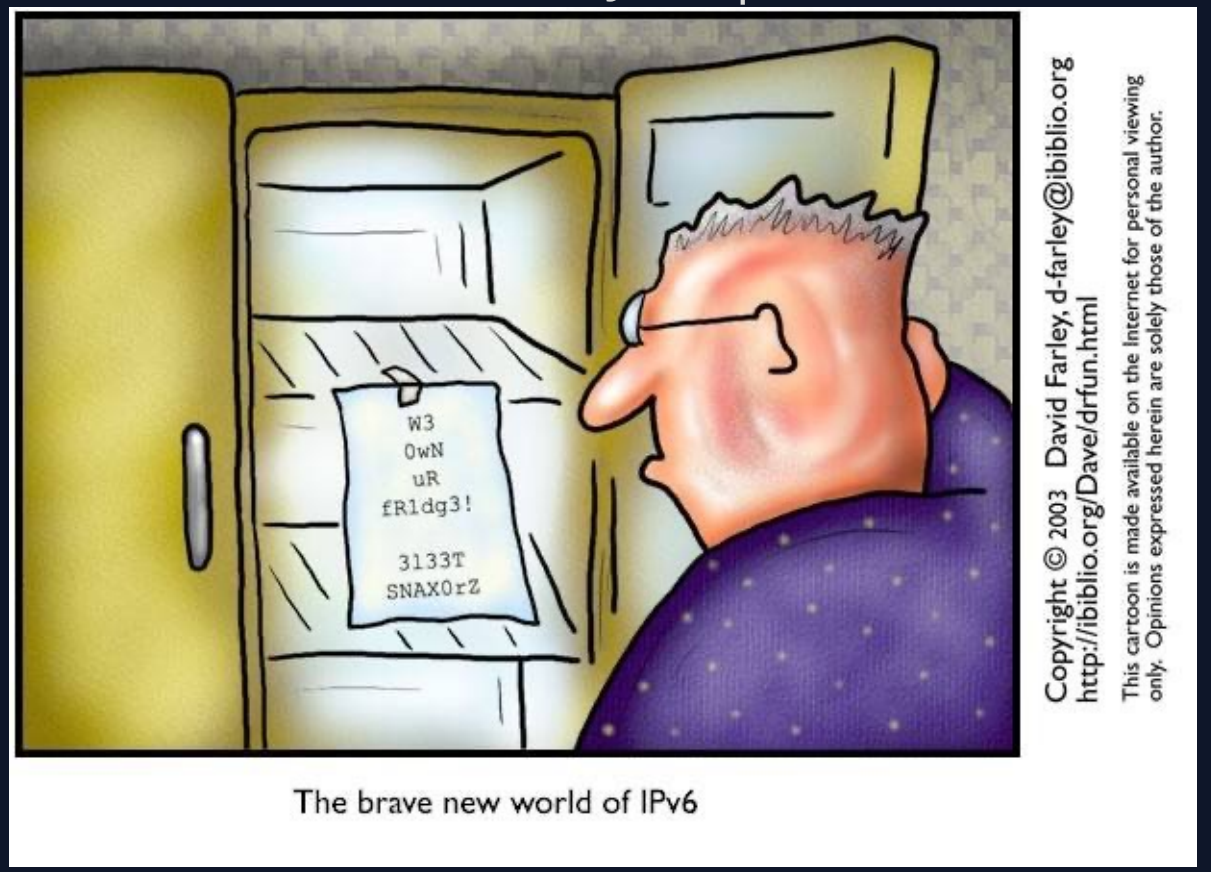
Network Layer: IPv4 vs IPv6 security

Feature	IPv4	IPv6
Address Space	32-bit, IP address exhaustion, easy to scan	128-bit, vast space reduces scanning attacks
IPsec Support	Optional, requires additional configuration	Native support, standardized implementation
Header Structure	Complex, contains optional fields	Simplified, fixed base header, extension hdrs
Broadcasting	Supports broadcasting (e.g., ARP, Smurf attacks)	Broadcast eliminated, relies on multicast
Neighbour Discovery	Uses ARP, vulnerable to spoofing and DoS	NDP w/ Secure Neighbour Discovery (SEND)
Routing Security	No integrated support for routing security	Supports route optimization, secure NDP
Privacy Extensions	Not natively supported	Privacy extensions prevent tracking
Extension Headers	Limited support, often requires workarounds	Allows flexible extension without modifying base header
Multicast and Anycast	Limited multicast support, anycast uncommon	Multicast and anycast built-in and standardized

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Network Layer: IPv6 security

- However, IPv6 also introduces complexity in deployment, especially with extension headers and NDP, which may require additional security.



The brave new world of IPv6

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Network Layer: IPv6 security

Extension Headers Manipulation and Fragmentation Attacks

- IPv6 EHs allow to add optional information, for flexible packet handling
- headers such as Destination Options or Routing Headers can create processing overhead, leading to potential DoS attacks
- attackers can exploit EH by:
  - crafting packets with unusual header chains to bypass filtering rules in firewalls and IDS/IPS
  - attacker could use Routing Header Type 0 (RH0), which specifies that packets should be routed through multiple intermediate nodes ("ping-pong" DoS amplification attack)
  - RH0 currently deprecated (RFC 5095)
- **Mitigation:**
  - firewalls and IDS/IPS to inspect and handle IPv6 EH correctly, blocking headers known for evasion tactics
  - network devices* must be configured to discard fragmented packets with dangerous header types (ensuring that all fragments of a packet are are received before forwarding)

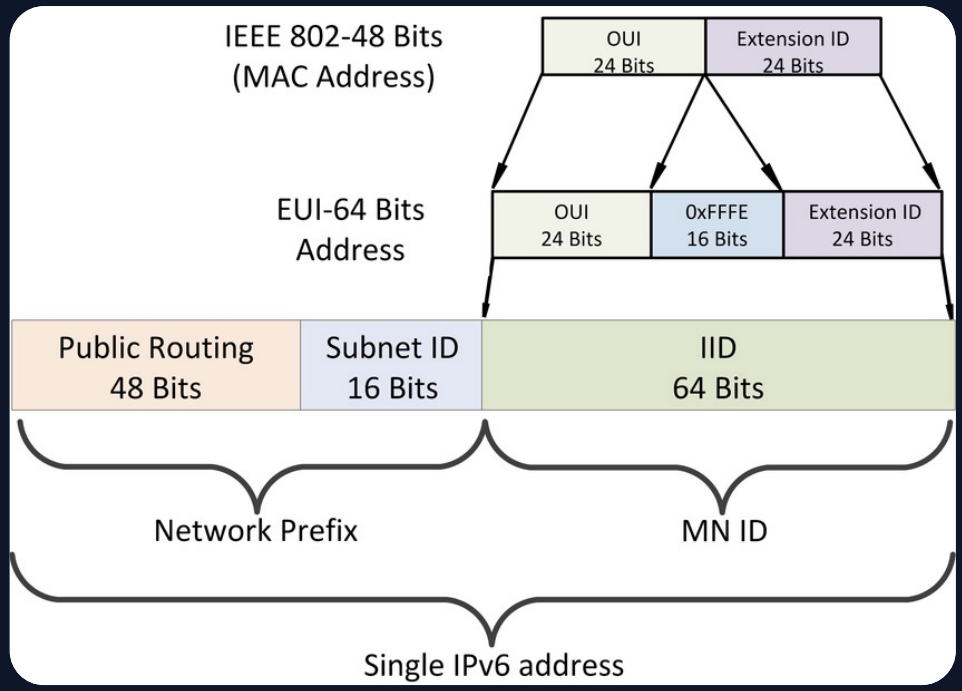
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* Usually firewalls and IDS/IPS

Network Layer: IPv6 security

Addressing and Privacy

- IPv6 allows for a large address space, including both link-local addresses (for local networks) and global unicast addresses
- MAC-derived IPv6 address (EUI-64 format)* reveals information about the hardware of the device, compromising anonymity
- **Mitigation:**
  - use Temporary Addresses (RFC 8981), randomizing IID bits frequently
  - use Stable Privacy Addresses (RFC 7217) where address is not related on MAC address (not hardware- but hashed-based)



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* Extended Unique Identifier Addresses (also used in FireWire (IEEE 1394), Zigbee, BLE Low , and InfiniBand)

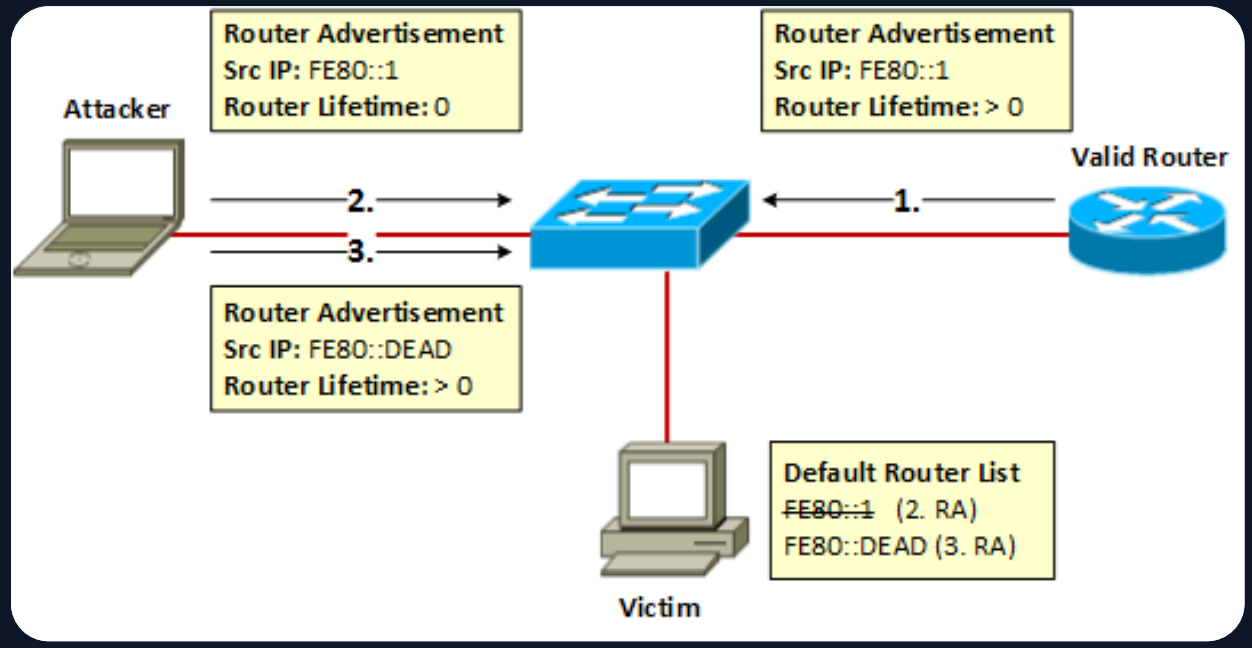
Network Layer: IPv6 security

Neighbor Discovery Protocol (NDP)

- NDP replaces ARP in IPv6, handling essential functions like address resolution, router discovery, and reachability detection
- NDP is vulnerable to attacks similar to ARP spoofing, such as ND spoofing and router advertisement (RA) flooding, leading to DoS or MitM attacks

Mitigation:

- RA guard and dynamic host configuration protections on network devices can also help limit RA flooding
- implement SEND (SEND adoption is currently limited due to its complexity and lack of universal support) to validate IPv6 src



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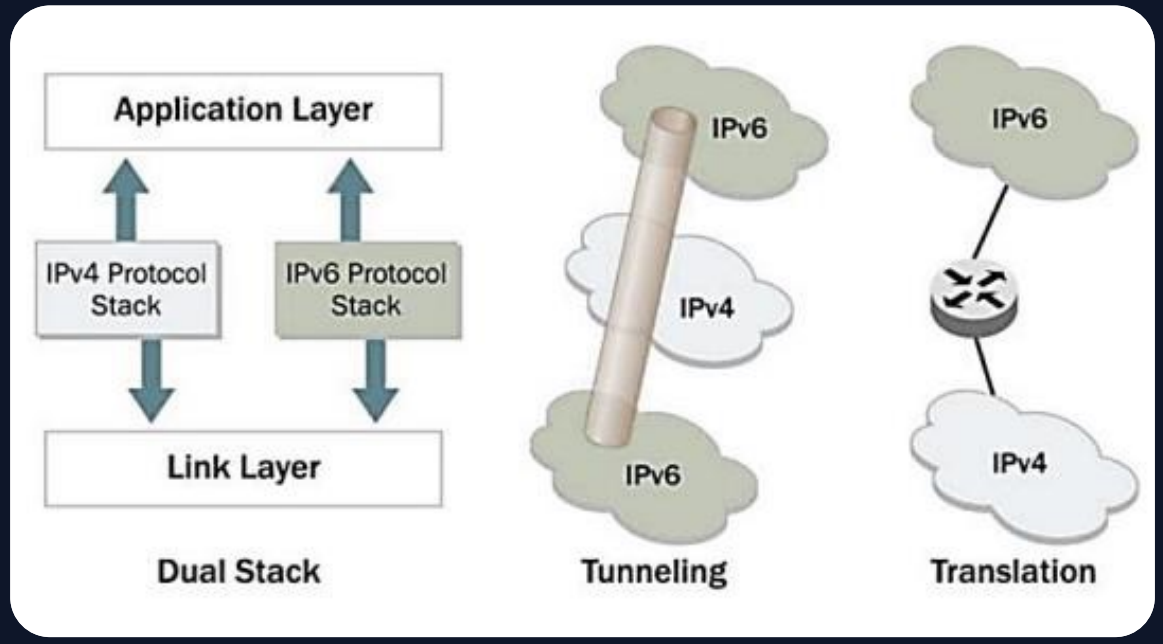
Network Layer: IPv6 security

IPv4 ↔️ IPv6

- IPv4 and IPv6 interoperability relies on various transition mechanisms, including dual stack, tunnelling (e.g., 6to4, ISATAP, Teredo) or translation
- transition mechanisms create additional attack surfaces, e.g., tunnelling protocols can be used to bypass firewall policies and encapsulate malicious IPv6 traffic over IPv4

Mitigation:

- avoid using insecure tunnelling protocols, prefer native IPv6 deployments whenever possible
- disable unused transition services, such as Teredo or 6to4, if not required



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Quizz